

Expo Cloud Deployment & Developer Handbook

This document outlines how to move your local project to the Expo Cloud so you can run it on any mobile device without needing your laptop, along with essential commands for React Native development.

Part 1: Running Projects Without a Laptop (EAS Update)

By following these steps, you "publish" your JavaScript bundle to Expo's servers. Your phone will download this bundle whenever you open the app in Expo Go.

1. Installation & Authentication

First, ensure the EAS (Expo Application Services) CLI is installed globally on your machine.

```
# Install EAS CLI
npm install -g eas-cli

# Login to your Expo account
eas login
```

2. Project Configuration

This step links your local folder to a project ID on your Expo Dashboard.

```
# Run this inside your project root
eas update:configure
```

3. Publishing to the Cloud

Whenever you want to "upload" your latest code so it's accessible from your phone:

```
# This sends your code to the 'production' branch on Expo's servers
eas update --branch production --message "Initial cloud version"
```

4. Accessing on Mobile

1. Download **Expo Go** from the App Store (iOS) or Play Store (Android).
2. Log in with the same account.

3. Navigate to the "**Projects**" tab.
4. Tap your project name to launch it.

Part 2: Essential Developer Commands

Project Initialization & Start

Command	Purpose
<code>npx create-expo-app@latest ProjectName</code>	Create a new project
<code>npx expo start</code>	Start the development server
<code>npx expo start -c</code>	Start and clear the cache (fixes 90% of glitches)
<code>npx expo start --ios</code>	Start and open iOS simulator directly
<code>npx expo start --android</code>	Start and open Android emulator directly

Library Management

Always use `npx expo install` instead of `npm install` for React Native packages. This ensures version compatibility with your Expo SDK.

```
# Standard install
npx expo install react-native-reanimated

# Uninstalling
npm uninstall react-native-reanimated

# Check for library issues
npx expo install --check
```

Troubleshooting & Cleanup

If your app is crashing or behaving strangely after changing versions:

```
# Remove build artifacts
rm -rf node_modules
rm -rf .expo
rm package-lock.json

# Fresh install
```

```
npm install
```

Navigation & State (Common Logic)

- **Navigate to a screen:** `navigation.navigate('ScreenName', { id: 123 })`
- **Get params:** `const { id } = route.params;`
- **Go back:** `navigation.goBack();`

Part 3: Advanced Cloud Features (Summary)

EAS Build (Standalone Apps)

If you want a real `.apk` or `.ipa` file that doesn't need Expo Go:

```
# Build for Android (generates a link to download the APK)
eas build --platform android --profile preview
```

Secret Management

If your app uses API keys (like Google Maps) that you don't want in your code:

```
# Add a secret to the cloud
eas secret:create --name API_KEY --value your_actual_key
```

Pro-Tips for Success

1. **Always Restart after EAS Update:** Once you update, your phone might take 5-10 seconds to "see" the new update. Force close Expo Go and reopen it to trigger the check.
2. **Matching SDKs:** If you update your laptop's Expo version, make sure your phone's **Expo Go** app is also updated from the App Store.
3. **Environment Variables:** Remember that `eas update` only publishes your code. If you use a `.env` file, ensure those variables are handled via `app.config.js`.